

Shankar

10.3.2

$$|\psi\rangle = \frac{1}{\sqrt{3}} \{ |3,3,4\rangle + |3,4,3\rangle + |4,3,3\rangle \}$$

Liboff

12.20

$$\hat{\chi}_{\pm} = \frac{1}{\sqrt{2}} \{ 1 \pm \hat{\chi} \}$$

$$\hat{\chi}_{+} \phi(1,2) = \frac{1}{\sqrt{2}} (\phi(1,2) + \phi(2,1)) = \phi_s(1,2)$$

$$\hat{\chi}_{-} \phi(1,2) = \frac{1}{\sqrt{2}} (\phi(1,2) - \phi(2,1)) = \phi_A(1,2)$$

11.73

a)

$$\hat{\rho} = \frac{1}{2} |E_1\rangle\langle E_1| + \frac{1}{2} |E_5\rangle\langle E_5|$$

$$j(x|E_i) = \sqrt{\frac{2}{L}} \sin\left(\frac{i\pi}{L}x\right)$$

$$b) |\psi_1\rangle = \frac{1}{\sqrt{2}} |E_1\rangle + \frac{1}{\sqrt{2}} |E_5\rangle$$

$$|\psi_2\rangle = \frac{1}{\sqrt{2}} |E_1\rangle - \frac{1}{\sqrt{2}} |E_5\rangle$$

$$\hat{\rho} = \frac{1}{2} |\psi_1\rangle\langle\psi_1| + \frac{1}{2} |\psi_2\rangle\langle\psi_2|$$

$$= \frac{1}{2} |E_1\rangle\langle E_1| + \frac{1}{2} |E_5\rangle\langle E_5|$$

② 10.3.5

(part 1) Let $|p\rangle$ be an eigenstate of $\hat{P}_{12}$ with eigenvalue p .

$$\hat{P}_{12}|p\rangle = p|p\rangle$$

$$\hat{P}_{12}^2|p\rangle = p^2|p\rangle \quad \text{but } \hat{P}_{12}^2 = \mathbb{1}, \quad \text{so } p^2 = 1$$

$$\Rightarrow \underline{p = \pm 1.}$$

(part 2) Expand $|w_1, w_2\rangle$ in $|x_1, x_2\rangle$ basis:

$$|w_1, w_2\rangle = \int dx_1 \int dx_2 \langle x_1, x_2 | w_1, w_2 \rangle |x_1, x_2\rangle = \int dx_1 dx_2 w_{12}(x_1, x_2) |x_1, x_2\rangle$$

$$\hat{P}_{12}|w_1, w_2\rangle = \int dx_1 \int dx_2 w_{12}(x_1, x_2) |x_2, x_1\rangle$$

Rename integration variables $x_1 \leftrightarrow x_2$

$$= \int dx_2 \int dx_1 w_{12}(x_2, x_1) |x_1, x_2\rangle$$

$$= \int dx_2 \int dx_1 \langle x_2, x_1 | w_1, w_2 \rangle |x_1, x_2\rangle$$

$$\text{but } \langle x_2, x_1 | w_1, w_2 \rangle = \left(\langle x_2 |^{(1)} \otimes \langle x_1 |^{(2)} \right) \left(|w_1\rangle^{(1)} \otimes |w_2\rangle^{(2)} \right)$$

$$= \langle x_2 | w_1 \rangle \langle x_1 | w_2 \rangle$$

$$= \langle x_1 | w_2 \rangle \langle x_2 | w_1 \rangle$$

$$= \langle x_1, x_2 | w_2, w_1 \rangle$$

$$\text{So } \hat{P}_{12}|w_1, w_2\rangle = \int dx_2 \int dx_1 \langle x_1, x_2 | w_2, w_1 \rangle |x_1, x_2\rangle = |w_2, w_1\rangle$$

(part 3) Show identities by using action on $|\psi\rangle$:

$$\begin{aligned}
 \hat{P}_{12} \hat{X}_1 \hat{P}_{12} |\psi\rangle &= \int dx_1 \int dx_2 \hat{P}_{12} \hat{X}_1 \hat{P}_{12} |x_1 x_2\rangle \langle x_1 x_2 | \psi \rangle \\
 &= \int dx_1 \int dx_2 \hat{P}_{12} \hat{X}_1 \hat{P}_{12} |x_1 x_2\rangle \psi(x_1, x_2) \\
 &= \int dx_1 \int dx_2 \hat{P}_{12} \hat{X}_1 |x_2 x_1\rangle \psi(x_1, x_2) \\
 &= \int dx_1 \int dx_2 x_2 \hat{P}_{12} |x_2 x_1\rangle \psi(x_1, x_2) \\
 &= \int dx_1 \int dx_2 x_2 |x_1 x_2\rangle \psi(x_1, x_2) \\
 &= \int dx_1 \int dx_2 \hat{X}_2 |x_1 x_2\rangle \psi(x_1, x_2) \\
 &= \int dx_1 \int dx_2 \hat{X}_2 |x_1 x_2\rangle \langle x_1 x_2 | \psi \rangle \\
 &= \hat{X}_2 |\psi\rangle
 \end{aligned}$$

To show for $\hat{P}_{12} \hat{X}_2 \hat{P}_{12} = \hat{X}_1$, switch $1 \leftrightarrow 2$ above.

To show for $\hat{P}_{12} \hat{P}_1 \hat{P}_{12} = \hat{P}_2$, show as above but expand in p_1, p_2 eigenstates instead.

$$\text{Assume } \hat{\Omega} = \sum_{mnkl} a_{mnkl} \hat{X}_1^m \hat{X}_2^n \hat{P}_1^k \hat{P}_2^l$$

$$\hat{P}_{12} \hat{\Omega} \hat{P}_{12} = \sum_{mnkl} a_{mnkl} \hat{P}_{12} \hat{X}_1^m \hat{X}_2^n \hat{P}_1^k \hat{P}_2^l \hat{P}_{12}$$

Now insert $\hat{P}_{12} \hat{P}_{12} = 1$ between each pair of operators:

$$\begin{aligned}
 &= \sum_{mnkl} a_{mnkl} \underbrace{\hat{P}_{12} \hat{X}_1^m \hat{P}_{12}}_{= (\hat{P}_{12} \hat{X}_1 \hat{P}_{12})^m} \underbrace{\hat{P}_{12} \hat{X}_2^n \hat{P}_{12}}_{= (\hat{P}_{12} \hat{X}_2 \hat{P}_{12})^n} \underbrace{\hat{P}_{12} \hat{P}_1^k \hat{P}_{12}}_{= (\hat{P}_{12} \hat{P}_1 \hat{P}_{12})^k} \underbrace{\hat{P}_{12} \hat{P}_2^l \hat{P}_{12}}_{= (\hat{P}_{12} \hat{P}_2 \hat{P}_{12})^l} \\
 &= (\hat{P}_{12} \hat{X}_1 \hat{P}_{12})^m (\hat{P}_{12} \hat{X}_2 \hat{P}_{12})^n (\hat{P}_{12} \hat{P}_1 \hat{P}_{12})^k (\hat{P}_{12} \hat{P}_2 \hat{P}_{12})^l = (\hat{P}_{12} \hat{\Omega} \hat{P}_{12})^m = \hat{X}_1^m
 \end{aligned}$$

$$\begin{aligned} \text{so } \hat{P}_{12} \hat{\Omega} \hat{P}_{21} &= \sum_{mkl} a_{mkl} X_2^m X_1^n P_2^k P_1^l \\ &= \hat{\Omega}(\hat{X}_2, \hat{X}_1, \hat{P}_2, \hat{P}_1) \quad \text{QED.} \end{aligned}$$

(part 4) $\hat{H} = T_1 + T_2 + \underbrace{V(x_1, x_2)}_{= V(x_2, x_1)} \text{ because particles identical.}$

$$\begin{aligned} \text{so } \hat{P}_{12} \hat{H} \hat{P}_{12} &= \hat{P}_{12} (\hat{T}_1 + \hat{T}_2 + \hat{V}(x_1, x_2)) \hat{P}_{12} \\ &= \hat{T}_2 + \hat{T}_1 + \hat{V}(x_2, x_1) = \hat{H}. \end{aligned}$$

$$\begin{aligned} \hat{U} &= e^{\frac{-i\hat{H}t}{\hbar}} = \sum_m \frac{1}{m!} \left(\frac{-i\hat{H}t}{\hbar} \right)^m \\ \text{so } \hat{P}_{12} \hat{U} \hat{P}_{12} &= \sum_m \frac{1}{m!} \left(\frac{-i\hat{H}t}{\hbar} \right)^m \underbrace{\hat{P}_{12} \hat{H}^m \hat{P}_{12}}_{= \hat{H}^m} \\ &= \sum_m \frac{1}{m!} \left(\frac{-i\hat{H}t}{\hbar} \right)^m = \hat{U}. \end{aligned}$$

Now take $|P_{12}^\pm\rangle$ such that $\hat{P}_{12} |P_{12}^\pm\rangle = \pm |P_{12}^\pm\rangle$

$$\begin{aligned} \hat{P}_{12} |P_{12}^\pm(t)\rangle &= \hat{P}_{12} \hat{U} |P_{12}^\pm(0)\rangle \\ &= \hat{P}_{12} \hat{P}_{12} \hat{U} \hat{P}_{12} |P_{12}^\pm(0)\rangle \\ &= \hat{U} \hat{P}_{12} |P_{12}^\pm(0)\rangle \\ &= \pm \hat{U} |P_{12}^\pm(0)\rangle \\ &= \pm |P_{12}^\pm(t)\rangle. \end{aligned}$$

④ $\hat{\rho} = \sum_i w_i |i\rangle\langle i| \rightarrow$ evolve $|i\rangle, \langle i|$ in time.

$$\text{so } \hat{\rho}(t) = \sum_i w_i \hat{U}(t) |i\rangle\langle i| \hat{U}^\dagger(t)$$

$$= \hat{U}(t) \sum_i w_i |i\rangle\langle i| \hat{U}^\dagger(t)$$

$$= \hat{U}(t) \hat{\rho}(0) \hat{U}^\dagger(t)$$

⑤ A pure ensemble has one $w_i = 1$, all others 0.

$$\text{So } \hat{\rho}(0) = |i\rangle\langle i|; \hat{\rho}^2 = |i\rangle\langle i| \hat{\rho} = |i\rangle\langle i| = \hat{\rho}$$

$$\hat{\rho}(t) = \hat{U}(t) |i\rangle\langle i| \hat{U}^\dagger(t)$$

$$\hat{\rho}^2(t) = \hat{U}(t) |i\rangle\langle i| \hat{U}^\dagger(t) \hat{U}(t) |i\rangle\langle i| \hat{U}^\dagger(t)$$

$$= \hat{U}(t) |i\rangle\langle i| \hat{U}^\dagger(t)$$

$$= \hat{U}(t) |i\rangle\langle i| \hat{U}^\dagger(t) = \hat{\rho}(t)$$

so if $\hat{\rho}^2(0) = \hat{\rho}(0)$, then $\hat{\rho}^2(t) = \hat{\rho}(t)$.